Upside Down Inflation: dark matter and reheating after a CPT-symmetric bang

B. H. Wiseman

Linnet Labs, Sydney, Australia (independent researcher; no external funding)

Abstract

Boyle, Finn and Turok proposed that the universe before the big bang is the CPT image of the one after it, with no inflation, and that the dark matter is a right-handed neutrino N_1 made by the expansion at the bang. Without inflation its mass is at most 491.6 PeV for every CPT-invariant state [1]. Here we add a period of inflation after the bang on each side. CPT symmetry makes a homogeneous inflaton even in conformal time, so it starts at rest. With enough inflation to make the observed perturbations, the choice of state at the bang drops out under the standard assumption that modes far above the temperature start in their vacuum, and the reheating history sets the mass, as for any purely gravitational Majorana fermion, provided the inflaton's Yukawa coupling to N_1 is below about $M/20M_P$. We also locate where an analytic spectrum for this transition departs from numerical ones. The inflaton must also survive the hot era before inflation and still reheat. A T-model α-attractor with $\alpha = 0.0045$ supplies the perturbations at $H_{\rm inf} = 10^{12}$ GeV, with $r = 1.6 \times 10^{-5}$ and $n_s = 0.9655$. Coupled to matter through its bounded field, it is shielded from the plasma before inflation, which moves it by $2.0 \times 10^{-5} M_P$. A Yukawa coupling to a heavy fermion reheats it to at most 1.2×10^{13} GeV, which puts the dark matter at 561 PeV or heavier. A trilinear coupling to the Higgs through the same field reaches instant reheating with the plateau intact, but meets tachyonic preheating. We close with a hypothesis, kept apart from the rest: that expansion counts the formation of clocks.

1. Introduction

Boyle, Finn and Turok proposed that the universe before the big bang is the CPT image of the universe after it [2,3]. Near the bang the scale factor grows linearly in conformal time, the radiation era runs back to the bang itself, and the dark matter is a right-handed neutrino N_1, kept stable by a $\mathbb{Z}_2$ and made by the expansion, with a mass near 480 PeV. Their universe needs no inflation, though they mention an inflationary neck at the bang as an option [3]. A companion paper showed that without inflation no CPT-invariant state makes fewer N_1 than theirs, so that 491.6 PeV is a ceiling on the mass, and corrected the rate at which graviton exchange in the plasma makes N_1 [1].

Others have put inflation into CPT-symmetric pairs of universes. Chaichian, Gogberashvili, Mnatsakanova and Tsiskaridze build a globally CPT-symmetric pair of coordinate-reversed universes, each with an inflaton, and let local CPT violation set the baryon asymmetry [4]; Kumar creates a flat universe-antiuniverse pair from nothing [5]. Here we keep Boyle, Finn and Turok's radiation bang and add a period of inflation after it on each side, so that each side, read with its own arrow of time pointing away from the bang, has the ordinary history of radiation, inflation, reheating and radiation again.

Section 2 shows that the symmetry starts the inflaton at rest and sets out the history. Section 3 follows the dark matter through it: the choice of state drops out, the reheating history sets the mass, and we locate the step at which an analytic spectrum for the transition [6] departs from the numerical ones [7], a disagreement its authors left open. That result assumes an inflaton that rests on its plateau at the bang, survives the hot radiation era before inflation and still reheats. Section 4 shows that couplings through the inflaton itself fail the middle requirement and builds an inflaton that meets all three. Section 5 states a hypothesis, kept apart from the rest, about what drives expansion.

2. The inflaton at the bang

2.1 CPT starts it at rest

Near the bang $a(\eta) = a_1\eta$, and Boyle, Finn and Turok relate the two sides by $(\eta, \mathbf{x}) \mapsto (-\eta, -\mathbf{x})$ together with charge conjugation [2]. A real scalar goes to $\phi(-\eta, -\mathbf{x})$ under this map, so a homogeneous inflaton on a CPT-symmetric background obeys $\phi(\eta) = \phi(-\eta)$. Its equation of motion near the bang,

$$\phi'' + \frac{2}{\eta}\,\phi' + a_1^2\eta^2\,V'(\phi) = 0,$$

has one solution regular at the bang and one that diverges as $1/\eta$. The regular one is even,

$$\phi = \phi_0 - \frac{a_1^2 V'(\phi_0)}{20}\,\eta^4 + O(\eta^8),$$

so the inflaton is uniform and at rest at the bang, at whatever value ϕ_0 it has there. Regularity alone picks this solution on each side; CPT symmetry then makes the other side start from the same ϕ_0, at rest too. The symmetry does not fix ϕ_0; inflation needs it on the plateau, far enough out to give the observed number of e-folds.

2.2 History

We take inflation to last 58 e-folds or more, so that it also makes the observed perturbations: today's Hubble scale left the horizon 58 e-folds before the end for $H_{\rm inf} = 10^{12}$ GeV and $T_{\rm rh} = 1.87\times10^{13}$ GeV. The bang then lies far outside the observable universe, and the inflaton supplies the scalar amplitude, which at $H_{\rm inf} = 10^{12}$ GeV needs a slow-roll parameter $\epsilon = 1.0\times10^{-6}$ and gives $r = 1.6\times10^{-5}$.

Inflation removes what the plasma made before it. If it runs up to the reduced Planck temperature, the plasma before inflation makes N_1 with yield 2.5×10^{-5}, 2.8×10^{13} times the dark matter, and inflation divides it by at least e^{3N}, so 12.6 e-folds bring it below a thousandth of the dark matter. After inflation the plasma starts at $T_{\rm rh}$, so its yield is bounded, and the expansion's can match it.

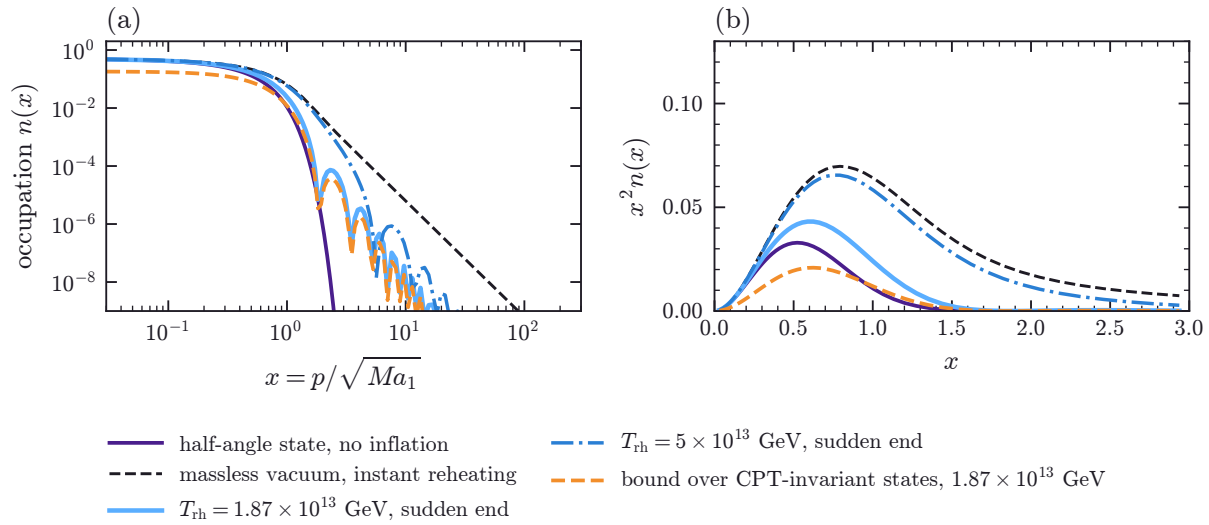
3. Dark matter after inflation

3.1 The state at the end of inflation

Near a radiation bang, $a = a_1\eta$, each mode of N_1 with comoving momentum p crosses a two-level gap, a Landau-Zener problem in $t = \sqrt{Ma_1}\,\eta$ with gap $2x$, $x = p/\sqrt{Ma_1}$. CPT invariance leaves a family of states at the bang, with one free phase for each momentum, and Boyle, Finn and Turok choose its least-excited member, the half-angle state; the in-vacuum is the adiabatic ground state long before the crossing. The abundance depends on the state through $I = \pi^{-1/2}\int_0^\infty x^2 n(x)\,dx$, with n the occupation after production, and the mass goes as $I^{-2/5}$ [1,3]. With a_1 the slope of the radiation era after reheating, the modes that become dark matter have $x \sim 1$. At the bang the same modes lie far above the comoving temperature of the plasma before inflation: their crossing variable is larger by a factor $e^N(\rho_{\rm end}/\rho_{\rm rh})^{1/12}$, which exceeds 10^6 already at $N = 12.6$, with $\rho_{\rm end} = 3H_{\rm inf}^2M_P^2$ and $\rho_{\rm rh}$ the radiation density at $T_{\rm rh}$. For these modes the bang's crossing is adiabatic, and the in-vacuum, the out-vacuum and Boyle, Finn and Turok's half-angle state differ from the massless vacuum by populations of order $1/(16x^4)$, 5×10^{-26} at $x = 1.06\times10^6$. During inflation the modes stay relativistic, $Ma \ll p$, so the field is conformal and inflation neither excites nor relaxes them. Boyle, Finn and Turok's state, which is Hadamard [8], therefore reaches the end of inflation as the massless vacuum on these modes, and the particles are made afterwards, once H falls to M.

Other members of the family differ from Boyle, Finn and Turok's state by populations of these modes, and quanta of these modes are energetic: their momentum is $\kappa\, x$ times the temperature, with κ about 10^{22} for 60 e-folds, so even at 8.4×10^{14} GeV, where inflation starts, it lies far above the Planck scale. We make the standard assumption that such trans-Planckian modes start in their vacuum, imposed here as the condition that N_1 carries less energy than the plasma before inflation [9,10]. The CPT symmetry plays no part in this step; the assumption removes the choice of state from the calculation. Appendix A shows how much it removes. Two half-crossings bracket the bang, ours and the mirror's, and their interference allows states that make about half as many particles as the massless vacuum, but any population that changes production by ten per cent carries more N_1 energy than the plasma once inflation lasts 12.7 to 14.1 e-folds (Table A.1), and at 58 e-folds 10^{78} times as much. Below about 12 e-folds nothing fixes the state.

Figure 1. After inflation the dark-matter modes start the radiation era in the massless vacuum, and the end of inflation and reheating shape their occupation. (a) Occupation after production and (b) the integrand $x^2 n$ of the production integral, against $x = p/\sqrt{M a_1}$ with a_1 the slope of the radiation era after reheating, for $H_{\rm inf} = 10^{12}$ GeV. Black dashed: the massless vacuum, which instant reheating gives, falling as $1/16x^4$. Blue: a sudden end of reheating at $T_{\rm rh} = 1.87 \times 10^{13}$ (solid) and 5×10^{13} GeV (dash-dotted), whose junctions cause the fringes; the end of inflation steepens the tail to x^{-6} beyond $x \sim 1/t_{\rm rh}$. Orange dashed: the bound $\bar{n}_*$ of Appendix A at 1.87×10^{13} GeV, averaged over the interference phase. Purple: the half-angle state of the universe without inflation, for comparison.



3.2 Production after inflation

From here the calculation is the standard one for gravitational production of a stable Majorana fermion after inflation [6,7,11-16], and Boyle, Finn and Turok noted that inflation turns their mass into a range [3]. We follow each mode through the end of inflation, reheating and the radiation era (Appendix A, Fig. 1). For reheating we take the inflaton to decay at a constant rate Γ, with $T_{\rm rh}$ defined by $\Gamma = H(T_{\rm rh})$ in the radiation era, and we integrate the graviton-exchange yield of the companion paper [1] along the temperature history, which rises above $T_{\rm rh}$ during reheating, to 7.7×10^{13} GeV when $T_{\rm rh} = 1.87 \times 10^{13}$ GeV. The mass solves $M\,(Y_{\rm exp} + Y_{\rm plasma}) = \rho_{\rm DM,0}/s_0$, with $Y_{\rm exp} \propto M^{3/2} I(M, T_{\rm rh})$, as Ema, Nakayama and Tang did for fermions and vector bosons [15].

The mass is set by the expansion when H falls to M. When that expansion is radiation-like, as for an inflaton oscillating in a quartic minimum, it is 261.8 PeV at every $T_{\rm rh}$ for $H_{\rm inf} = 10^{12}$ GeV. With radiation joined straight onto the end of inflation it is 269.4, 261.3 and 258.8 PeV for $H_{\rm inf} = 10^{11}$, 10^{12} and 10^{13} GeV, since the radiation-like era then begins where the mass term is already $(M/H_{\rm inf})^{1/2}$ in the variables of [1]. When a matter-like reheating from a quadratic minimum is still under way, the mass is higher: 449

PeV at $T_{\rm rh} = 1.87 \times 10^{13}$ GeV, 1112 PeV at 3×10^{12} GeV, and rising as $T_{\rm rh}^{-1/2}$ below, the scaling found for production during reheating [7,15]. Graviton exchange lowers the mass where the plasma is hot: for a quadratic minimum it makes 0.6 per cent of the abundance at 1.87×10^{13} GeV, 8 per cent at 5×10^{13} GeV and most of it above 1.05×10^{14} GeV. At $H_{\rm inf} = 10^{12}$ GeV the mass therefore runs from 4.2×10^5 GeV, when reheating is instant and the plasma at 8.4×10^{14} GeV sets the abundance, up to about $H_{\rm inf}$, which the $T_{\rm rh}^{-1/2}$ scaling reaches at $T_{\rm rh}$ near 3.7×10^6 GeV. There is no mass prediction; Table 1 and Fig. 2 give benchmarks.

The benchmarks depend on the reheating model. A sudden end of reheating and a perturbative decay differ by 3 to 10 per cent in the mass at the same $T_{\rm rh}$, between 1.2×10^{13} and 5×10^{13} GeV, with the expansion alone, and along a perturbative history the plasma's yield is 0.53 of the sudden formula at $T_{\rm rh}$, in line with Garcia et al. once their reheating temperature, 0.82 of ours for the same Γ, is converted [17]. With a sudden end, the scale of inflation moves the mass by at most 0.5 per cent between $H_{\rm inf} = 10^{11}$ and 10^{13} GeV, and at 10^{10} GeV it is 3 per cent higher at $T_{\rm rh} = 1.87 \times 10^{13}$ GeV and 10 per cent higher at 3×10^{12} GeV. For a quartic minimum the inflaton's mass falls as $1/a$, so its decay rate is not constant [18]; we take $T_{\rm rh}$ to be the radiation-era temperature at which H equals that rate. Decays to fermions slow down and trilinear decays to bosons speed up, which moves the temperature above which graviton exchange makes most of the dark matter from about 9×10^{13} GeV for a constant rate to 3.1×10^{13} and 1.1×10^{14} GeV respectively. The condensate can also fragment [19]. None of this changes the radiation-like mass of the expansion. The minimal Einstein-Cartan contact term [1,20] moves the crossover for a quadratic minimum from 1.05×10^{14} to 5.2×10^{13} GeV. Since other operators can only add to the plasma's yield, in Einstein gravity the masses above $T_{\rm rh} \approx 3 \times 10^{13}$ GeV are upper bounds.

Table 1. With inflation there is no mass prediction: the mass follows the reheating temperature and the shape of the inflaton's minimum. Benchmark N_1 masses against reheating temperature, for $H_{\rm inf} = 10^{12}$ GeV, rounded to 1 PeV. For perturbative reheating from a quadratic minimum, the second and third columns are the expansion alone and with graviton exchange in the plasma, and the fourth is the plasma's share of the abundance. The fifth is the mass with graviton exchange for a quartic minimum, where reheating is radiation-like (constant decay rate), and the sixth for a quadratic minimum with a sudden end of reheating.

$T_{\rm rh}$ (GeV)	expansion (PeV)	with plasma (PeV)	plasma share	quartic (PeV)	sudden end (PeV)
3×10^{12}	1112	1112	0.0%	262	1077
8×10^{12}	681	681	0.1%	262	659
1.2×10^{13}	558	557	0.2%	262	540
1.87×10^{13}	451	449	0.6%	261	413
2.5×10^{13}	396	393	1.3%	260	353
3.5×10^{13}	348	343	3.1%	257	312
5×10^{13}	316	304	8.0%	249	278
7×10^{13}	296	270	19.3%	230	240
1×10^{14}	283	220	45.2%	181	172
1.4×10^{14}	275	144	79.3%	107	86
2×10^{14}	270	63	97.2%	46	31

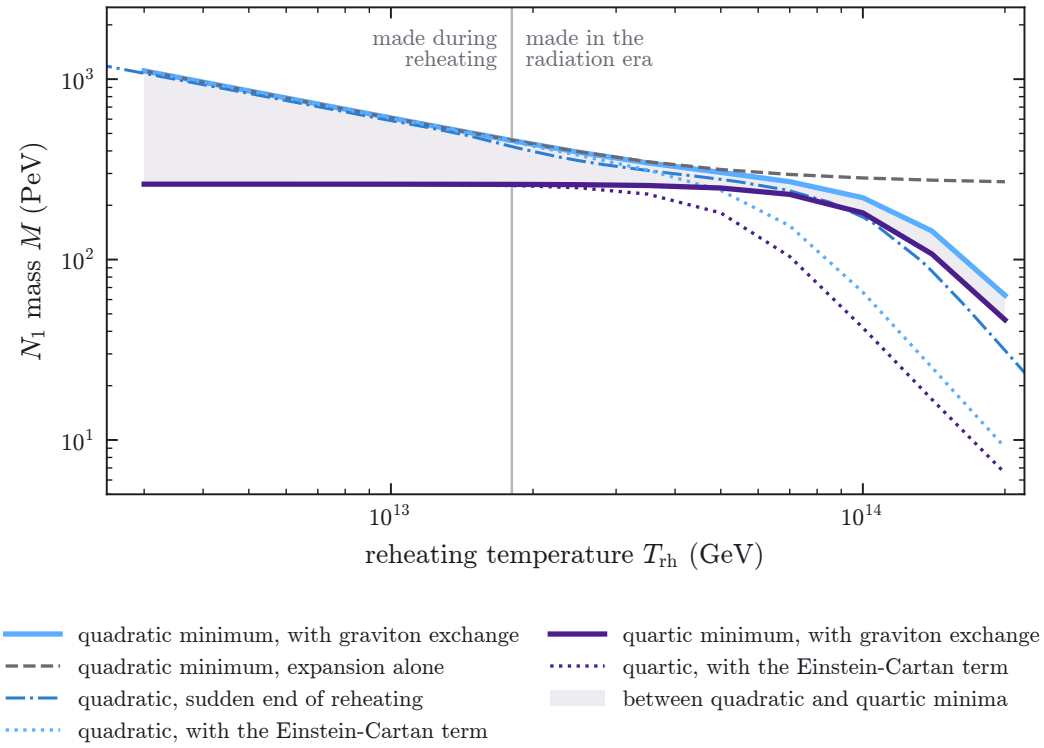


Figure 2. With inflation, the reheating temperature sets the dark-matter mass, and graviton exchange in the plasma lowers it when reheating is hot. Benchmark N_1 masses against reheating temperature, $H_{\rm inf} = 10^{12}$ GeV. Blue: perturbative reheating from a quadratic inflaton minimum, with graviton exchange in the plasma integrated along the temperature history; grey dashed: the same with the expansion alone; blue dash-dotted: a sudden end of reheating. Purple: a quartic minimum, where reheating is radiation-like; the shading spans the two minima. Dotted: the minimal Einstein-Cartan contact term added to graviton exchange, for the quadratic (blue) and quartic (purple) minima. The vertical line marks where $H(T_{\rm rh}) = M$ for a quadratic minimum; to its left the particles are made during reheating.

3.3 A published analytic spectrum

Our sudden limit agrees with the closest numerical calculation to within 15 per cent. In the form $na^3 = C_M M^3 a_M^3$, with a_M the scale factor at $H = M$, it gives $C_M = I/\pi^{3/2} = 0.0115$ for a Majorana fermion, against 0.010 from the numerical result of Chung, Everett, Yoo and Zhou for an abrupt transition from de Sitter to radiation, with the same k^{-4} tail, determined for $M < 0.1 H_{\rm inf}$; their 4×0.005 counts the four states of a Dirac fermion [7]. Beltrán-Palau, Nadal-Gisbert, Navarro-Salas and Pla give $I = 0.06305$ and 2.6×10^8 GeV for the massless-vacuum state [21], which is our instant-reheating limit; their integral is ours cut off near $x = 31$, where the slowly falling tail still carries 2 per cent.

Herring and Boyanovsky's analytic treatment of the same transition gives the half-angle occupation instead, $C_M = 0.0023$, with a Gaussian tail, and they noted that they did not understand the difference from Chung et al. [6]. Their matching is the same as ours, the massless spinor at the end of inflation projected onto the radiation era's out states at $z = 0$, and its exact evaluation gives

$$n_{\rm vac}(x) = \tfrac{1}{2}\Big[1 - \sqrt{1 - e^{-\pi x^2}}\,\cos\vartheta(x)\Big], \qquad \vartheta(x) = \tfrac{\pi}{4} + \arg\Gamma(\tfrac{1}{2} + \tfrac{ix^2}{4}) - \arg\Gamma(1 + \tfrac{ix^2}{4}),$$

which reproduces our mode integration to four figures, $1/(16x^4)$ tail included; exact solutions of Landau-Zener problems of this kind are in [22,23]. Their Appendix C (arXiv v2) replaces the Gamma-function

phases by their large-argument form, which sets $\vartheta \to 0$ and gives the half-angle formula, but $\vartheta(0) = \pi/4$ and $\vartheta(1) = 0.47$, so the approximation fails where the abundance is made; their printed Eq. (III.41) also carries the out-state normalisation once where the projection needs it squared. The difference with Chung et al. is that step.

3.4 Other couplings after inflation

The Higgs operator $(c_H/2\Lambda)\,\overline{N_1^c} N_1 H^\dagger H$ of the companion paper [1] gives a bounded yield after inflation, linear in $T_{\rm rh}$: with $\Lambda = M_P$ it stays below a tenth of the dark matter for $c_H < 4.1 \times 10^{-5}$ at $T_{\rm rh} = 1.87 \times 10^{13}$ GeV and $c_H < 2.6 \times 10^{-5}$ at 10^{14} GeV. Through a graviton the inflaton condensate can annihilate to N_1 pairs, but the amplitude needs a helicity flip and is suppressed by $(M/m_\phi)^2$ [24,25]; for a quadratic minimum Clery et al.'s rate gives under a tenth of a per cent of the dark matter for m_ϕ from 10^{13} to 3×10^{13} GeV and $H_{\rm inf}$ from 10^{12} to 10^{13} GeV; for a quartic minimum m_ϕ falls as $1/a$, so the suppression weakens. A singlet inflaton can also couple to N_1 pairs directly, $(y/2)\,\phi\,\overline{N_1^c} N_1$, which the $\mathbb{Z}_2$ allows. With $\Gamma(\phi \to N_1 N_1) = y^2 m_\phi/16\pi$, a coupling of the size the Majorana mass suggests, $y = M/M_P$, makes 40 times the dark matter at $T_{\rm rh} = 1.87 \times 10^{13}$ GeV, and a branching ratio above 3.5×10^{-20} ($m_\phi = 10^{13}$ GeV) makes more than a tenth of it. No symmetry of the model forbids the coupling while letting the inflaton decay to the Standard Model, so we assume $y < 9.2 \times 10^{-12}\,(T_{\rm rh}/1.87 \times 10^{13}\,{\rm GeV})^{1/2} (449\,{\rm PeV}/M)^{1/2}$, which does not depend on m_ϕ and is $0.05\,M/M_P$ at the reference point, just as we assumed the Higgs operator small.

4. An inflaton that survives the hot era

4.1 Couplings through the inflaton itself

The inflaton has to stay on its plateau through the radiation era before inflation, which begins near the Planck temperature, and still reheat the universe afterwards, though a coupling strong enough to reheat can move it off the plateau in that hot era. A Yukawa coupling to fermions that reheats to 1.87×10^{13} GeV from $m_\phi = 10^{13}$ GeV needs $y = 0.035$, and gives the inflaton a thermal mass of 1.2×10^{13} GeV, 12 times $H_{\rm inf}$, when the plasma at 8.4×10^{14} GeV meets the plateau, unless the fermions are heavier than the temperature there. The thermal mass exceeds the Hubble rate below 1.0×10^{16} GeV, so keeping the fermions heavier than the temperature needs $y\phi$ above that, and their one-loop energy, $(y\phi)^4/16\pi^2 \approx 6.9 \times 10^{61}$ GeV^4, would then exceed the plateau's 1.8×10^{61} GeV^4. A trilinear coupling to the Higgs barely moves the inflaton: at the largest coupling whose one-loop slope stays below the plateau's, the plasma shifts it by 0.4 per cent of M_P in total, and that coupling reheats perturbatively to at most about 3×10^{13} GeV, for $\phi \sim M_P$ and $m_\phi = 10^{13}$ GeV. Benchmarks with reheating above that temperature need an inflaton whose couplings are suppressed on the plateau.

The perturbative estimate leaves out preheating: after inflation the trilinear coupling makes the Higgs tachyonic for part of each oscillation, which can complete preheating within a few oscillations [26] and, outside a window of Higgs-inflaton couplings, destabilise the electroweak vacuum [27]. Section 3 used only the plateau's height and slope and the mass at the minimum. The rest of this section builds an inflaton with those properties whose couplings are suppressed on the plateau.

4.2 An α-attractor

The T-model α-attractor [28,29] has

$$V = V_0 \tanh^2 \frac{\phi}{\sqrt{6\alpha}\,M_P},$$

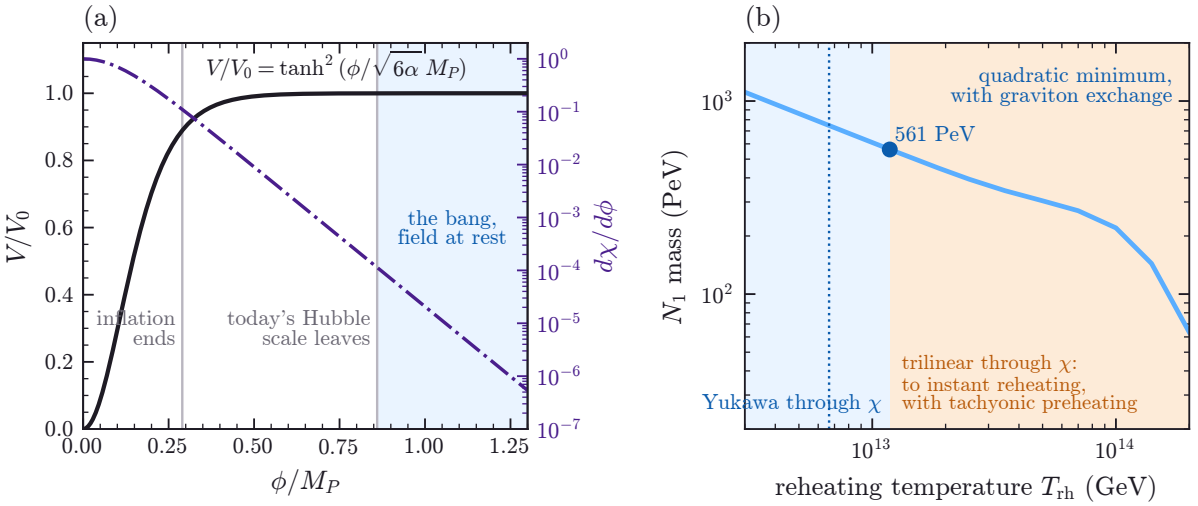
a plateau at large ϕ and a quadratic minimum at zero. In slow roll, with $u = \phi/\sqrt{6\alpha}\,M_P$, the first slow-roll parameter is $\epsilon = 4/(3\alpha \sinh^2 2u)$ and the number of e-folds to the end of inflation is $N = (3\alpha/4)(\cosh 2u -$

$\cosh 2u_{\rm end}$), inflation ending where $\epsilon = 1$. Section 2.2 needs $\epsilon = 1.0 \times 10^{-6}$ at 58 e-folds, which with the observed amplitude puts the plateau at $H_{\rm inf} = 10^{12}$ GeV. That fixes $\alpha = 0.0045$, and then $r = 1.6 \times 10^{-5}$, $n_s = 0.9655$, and the mass at the minimum is $m_\phi = H_{\rm inf}/\sqrt{\alpha} = 1.5 \times 10^{13}$ GeV. Today's Hubble scale leaves the horizon at $\phi = 0.86\, M_P$, and inflation ends at $0.29\, M_P$ (Fig. 3a).

The spectral index agrees with Planck's 0.965 ± 0.004 [30] and lies 2.8σ below the 0.974 ± 0.003 of ACT's combination with Planck, lensing and baryon acoustic oscillations [31]. If that value holds, α-attractors at 58 e-folds are disfavoured. Nothing below depends on the spectral index, only on the plateau's height and slope and the mass at the minimum, so an inflaton with a larger n_s at the same r would serve as well.

The plateau runs on without limit, so the bang may sit anywhere beyond $0.86\, M_P$. Between the bang and the start of inflation the plateau's slope moves the field only $1.6 \times 10^{-4} M_P$ from $0.86\, M_P$, 0.11 e-folds; the field waits while the radiation thins. Inflation starts when the radiation's energy falls to the plateau's, at a temperature of 8.4×10^{14} GeV.

Figure 3. (a) The T-model potential (black), with the field values where inflation ends and where today's Hubble scale leaves the horizon, and the range where the bang may sit with the field at rest (blue). Purple, right axis: $d\chi/d\phi$, the factor by which the bounded field shields the plateau from anything coupled to it. (b) The N_1 mass against the reheating temperature for a quadratic minimum, with graviton exchange in the plasma (blue line), as in Fig. 2. Blue: the reach of a Yukawa coupling through χ, to 1.2×10^{13} GeV (dot), and to 6.7×10^{12} GeV with the loop held to a tenth of V_0 (dotted). Orange: the further reach of a trilinear Higgs coupling through χ, up to instant reheating.



4.3 Shielded from the plasma

The plasma before inflation runs from near the Planck temperature down to 8.4×10^{14} GeV, and any coupling that later reheats the universe lets it push on the inflaton first. A Yukawa coupling through the inflaton itself, strong enough to reheat to 1.87×10^{13} GeV, gives the inflaton a thermal mass twelve times $H_{\rm inf}$ when the plasma meets the plateau (Section 4.1), which pulls the field off its plateau before inflation can start.

The α-attractor offers a way round this. In its superconformal construction the potential is a function of a field with a pole in its kinetic term [29], the bounded field

$$\chi = \sqrt{6\alpha}\, M_P \tanh \frac{\phi}{\sqrt{6\alpha}\, M_P},$$

and we let matter couple to the same field. On the plateau χ hardly changes: $d\chi/d\phi = \operatorname{sech}^2 u$ is 1.2×10^{-4}

at $0.86\,M_P$ and falls by e^{-2} for each further $\sqrt{6\alpha}\,M_P = 0.16\,M_P$ (Fig. 3a). A coupling $y\chi\bar{\psi}\psi$ to a heavy Dirac fermion ψ then pushes on the inflaton with

$$\frac{\partial F}{\partial \phi} = \frac{T^2}{\pi^2}\,K\Big(\frac{m}{T}\Big)\,2m\,\frac{dm}{d\phi}, \qquad m = y\chi, \qquad K(z) = \int_0^\infty \frac{x^2\,dx}{E\,(e^E+1)}, \quad E = \sqrt{x^2+z^2},$$

the derivative of the fermions' free energy, and $dm/d\phi = y\,\mathrm{sech}^2 u$ shields the push by the same factor. We follow the field from the reduced Planck temperature, at rest at $0.86\,M_P$, the start nearest the minimum, through the moment the plasma meets the plateau and on into inflation, once with this force and once without. At the largest coupling Section 4.4 allows, $y = 0.0182$, the plasma moves the field $2.0\times10^{-5}M_P$, which shortens inflation by 0.014 e-folds, and the curvature it adds never exceeds $1.7\times10^{-4}H^2$. The same coupling through ϕ itself, with the fermions light, drags the field from 0.86 to $0.08\,M_P$ before inflation could begin.

4.4 Reheating, and how hot

Near the minimum $\chi \approx \phi$, so the inflaton decays to ψ pairs at $\Gamma = y^2 m_\phi/8\pi$. Reheating ends when the Hubble rate of the radiation era falls to Γ, at $T_{\rm rh} = y\times 6.5\times10^{14}$ GeV, and ψ passes its energy on to the Standard Model. It can be, for example, a vector-like lepton with a small Yukawa coupling to the Higgs and a Standard Model lepton, through which it decays; that coupling keeps ψ in the plasma before inflation without touching the inflaton, whose own coupling stays shielded.

The coupling cannot be large, because ψ also changes the potential. Its one-loop potential [32], $-(4/64\pi^2)\,m^4[\ln(m^2/\mu^2) - 3/2]$ with $m = y\chi$, is $-c\,x^4\ln x^2$ in $x = \tanh(\phi/\sqrt{6\alpha}\,M_P)$, with $c = (y\sqrt{6\alpha}\,M_P)^4/16\pi^2$, plus a quartic that a counterterm absorbs. We fix the counterterms so that the plateau's height, which the observed amplitude sets, and the mass at the minimum keep their values; the loop then adds $c\,x^4|\ln x^2|$ to V_0x^2, zero at both ends. The slope of the sum is $2x[V_0 - c\,x^2(2\ln x^2+1)]$, and $x^2(2\ln x^2+1) \le 1$ for $x \le 1$, so the potential rises all the way to its plateau only while $c < V_0$. That allows y up to 0.0182, where $T_{\rm rh} = 1.2\times10^{13}$ GeV; holding c to a tenth of V_0 gives $y = 0.0102$ and $T_{\rm rh} = 6.7\times10^{12}$ GeV. This inflaton, with this coupling, reheats to at most 1.2×10^{13} GeV, below the reference point of Section 3, 1.87×10^{13} GeV.

These are perturbative estimates. Resonant production of the fermions can change them [33], but a fermionic channel gives the Higgs no tree-level coupling to the inflaton, so the tachyonic preheating that troubles trilinear couplings [26,27] does not arise.

Hotter reheating needs another coupling. The trilinear coupling to the Higgs of Section 4.1, moved onto the bounded field as $\mu\chi H^\dagger H$, loses the limit it had there, because its one-loop slope on the plateau is shielded by $\mathrm{sech}^2 u$ like everything else. Instant reheating, at 8.4×10^{14} GeV, needs $\mu = 1.9\times10^{13}$ GeV. There the Higgs loop's energy on the plateau is 3.2 per cent of the plateau's and its slope 2.1 per cent of the tree-level slope, and the plasma before inflation moves the field $7.2\times10^{-6}M_P$. What this coupling cannot escape is the oscillation after inflation: whenever $\chi < 0$ the Higgs is tachyonic, which can complete preheating within a few oscillations [26] and, outside a window of Higgs-inflaton couplings, destabilise the electroweak vacuum [27]. Whether that window holds the couplings needed here is the open question for hotter reheating.

4.5 The dark-matter mass

For a quadratic minimum and perturbative reheating, Table 1 gives the N_1 mass at each reheating temperature. This inflaton, reheating through ψ, gives 1112 PeV at 3×10^{12} GeV, 681 PeV at 8×10^{12} GeV, and 561 PeV at its limit of 1.2×10^{13} GeV (Fig. 3b), or 747 PeV with c held to a tenth of V_0. With this coupling the dark matter therefore weighs 561 PeV or more, above the 491.6 PeV that bounds it without inflation [1]. Lighter dark matter needs hotter reheating, which here means the trilinear route and its preheating question. With 58 or more e-folds, inflation predicts primordial tensor modes, $r = 1.6\times10^{-5}$ at $H_{\rm inf} = 10^{12}$ GeV, where the universe without inflation predicts none [3], but far below the reach of planned experiments.

5. A hypothesis: expansion as clock formation

Sections 2 to 4 are standard physics. This section states a hypothesis of the author's, and nothing in them depends on it.

The hypothesis is that the expansion of the universe counts the formation of clocks, physical records that order events in time, so that more clocks means more time and, with space and time joined, more space. It has a natural reading during inflation. Each mode of a light field that leaves the horizon freezes and decoheres into an effectively classical record of its amplitude [34,35], and in de Sitter space the rate at which that happens is fixed. Modes leave when their comoving wavenumber reaches aH. With $k^3/6\pi^2$ modes per comoving volume below k, the rate per physical volume is $H^4/2\pi^2$, which is $2/3\pi \approx 0.21$ new records per Hubble volume per Hubble time for each degree of freedom, whatever H is. A universe whose expansion is set by its rate of record formation would therefore expand steadily while records form steadily, which is what inflation does, and would stop when the supply of freezing modes runs out.

The reading is only consistent with inflation. Because $2/3\pi$ holds for every H, it does not fix the rate of inflation, and we have no law that turns a rate of record formation into a Hubble rate with a coefficient fixed elsewhere; until there is one, the hypothesis predicts nothing. One test exists, and it failed. In a late-time version the dark energy grows in proportion to the stars formed so far [36], with no free parameter. It raises the Hubble constant inferred from the cosmic microwave background by 9.8 per cent, but distances refuse it, by 83.5 in χ^2 against DESI's second-release baryon acoustic oscillations [37] and the Pantheon+ supernovae [38], and a free amplitude does not rescue it (an unpublished test, registered before it met the data). Inflation is a different regime, and the reading above is where the idea would have to start.

6. Conclusions

Inflation fits after a CPT-symmetric radiation bang. The symmetry starts a homogeneous inflaton at rest, and with inflation after the bang it no longer acts on the dark matter. Under the standard assumption that trans-Planckian modes start in their vacuum the choice of state drops out; at 58 e-folds a state that changed production by ten per cent would carry 10^{78} times the plasma's energy. As for any Majorana fermion coupled only through gravity, the reheating history sets the mass (Fig. 2, Table 1), from 4.2×10^5 GeV up to about $H_{\rm inf}$, and near 260 PeV when the expansion at $H = M$ is radiation-like. Along the way we located the step behind Herring and Boyanovsky's half-angle spectrum, which resolves their disagreement with Chung et al.

These results assume an inflaton that stays on its plateau through the hot era before inflation and still reheats, with its couplings suppressed on the plateau if reheating exceeds about 3×10^{13} GeV. They also assume that its Yukawa coupling to N_1 is at most a twentieth of what the Majorana mass suggests, about $M/20M_P$, and a Higgs operator no larger than that mass suggests. A T-model α-attractor with $\alpha = 0.0045$, coupled through its bounded field, sits out the hot era almost untouched, and a Yukawa coupling to a heavy fermion reheats it to 1.2×10^{13} GeV, which puts the dark matter at 561 PeV or more. Hotter reheating, and lighter dark matter, turn on whether a trilinear Higgs coupling can preheat without destabilising the electroweak vacuum. After inflation the CPT universe keeps its candidate and its massless lightest neutrino, and reheating sets the mass.

Appendix A. Modes through inflation

Numerics. `v6_reheating_production.py` integrates $i\partial_t\psi = (m(t)\sigma_z + x\sigma_x)\psi$ with an eighth-order Runge-Kutta method (relative tolerance 10^{-10}), starting each mode in the instantaneous ground state deep in inflation, at $t \le -5$ and where $m \le 0.02x$ and $m' \le 10^{-6}x^2$, and ending where $t \ge \max(40, 6x, 4t_{\rm rh} + 20)$, where it projects onto the adiabatic excited state. The condition on m' matters: the instantaneous ground state misses the evolved state by an amplitude of about $m'/2x^2$, which later interferes with the production amplitude, and without that condition the mass at $T_{\rm rh} = 3 \times 10^{12}$ GeV moves by 0.5 per cent for $H_{\rm inf} = 10^{11}$ GeV. A start where $m' \le 10^{-9}x^2$ changes I by under 10^{-6} (`v6_start_check.py`). For $x \le 6$ it uses 97 modes; beyond, first-order adiabatic perturbation theory gives $n = |\tilde{m}'(2x)|^2/4x^2$, with $\tilde{m}'$ the Fourier transform

of $m'(t)$, which we evaluate in closed form in the radiation and reheating segments and by oscillatory quadrature in the de Sitter one. The checks: a full sweep reproduces $P = e^{-\pi x^2}$ and $I_{\rm in} = 1/(4\pi^{3/2})$; the half-angle formula gives I_* to 10^{-10}; at $x = 6$ the first-order tail matches the exact solution within 5 per cent and the Fourier formula within 3 per cent for fast transitions; for slow transitions the tail beyond $x = 6$ carries under 1 per cent of I, a test the sudden limit, whose tail carries 9.1 per cent, fails as it should. `v6_mass_with_bath.py` solves for the mass with a sudden end of reheating on 38 reheating temperatures and four variants, and with `--ceiling` for the bound over all invariant states.

Perturbative reheating. `v6_reheating_smooth.py` and `v6_mass_smooth.py` solve $\dot\rho_\phi = -3H\rho_\phi - \Gamma\rho_\phi$ and $\dot\rho_r = -4H\rho_r + \Gamma\rho_\phi$ from the end of inflation, with de Sitter before it, read a_1 and the origin of conformal time off the late radiation era, and hand the mode solver $m(t) = \sqrt{M/a_1}\,a$ by spline interpolation. The graviton-exchange yield is integrated alongside, with $T^4 = 30\rho_r/\pi^2 g_*$. A decay within two Hubble times of the end of inflation reproduces the sudden model's mass to half a per cent, and its plasma yield is 0.84 of the sudden formula evaluated at the highest temperature reached; both are checks the code must pass. The perturbative grid covers 12 reheating temperatures.

Two half-crossings. Other members of the family behave differently from the massless vacuum. Two half-crossings bracket the bang, ours and the mirror's, separated by a long massless stretch in which the two massless levels acquire a relative phase 2Φ, with Φ of order $x\,e^N$. The massless vacuum gives the same late occupation $n_{\rm vac}(x)$ whatever Φ is. A state with population p_e in the upper massless level and a phase unrelated to Φ gives $n_{\rm vac} + p_e(1-2n_{\rm vac})$ once the rapid variation of Φ with x averages the interference away, so it makes more particles; a state whose phases follow Φ can make fewer. The companion paper's bound over CPT-invariant states [1] still applies to the whole evolution, and now it oscillates. For the in-vacuum $P = 4n_{\rm vac}(1-n_{\rm vac})\sin^2(\Phi+\phi_0)$, the Stückelberg interference of two passages [39], and $n_* = (1-\sqrt{1-P})/2$ averages over Φ to

$$\bar n_* = \tfrac12\Big[1 - \tfrac{2}{\pi}E(4n_{\rm vac}(1-n_{\rm vac}))\Big] \approx \tfrac12\, n_{\rm vac},$$

with E the complete elliptic integral of the second kind. Boyle, Finn and Turok's criterion, least particle number and energy, then has two readings: applied at the bang, where the symmetry is imposed, it picks the half-angle state of the bang's own crossing, which for these modes is the massless vacuum; applied to particle number at late times, it picks the state that reaches $\bar n_*$. Without inflation the two coincide. At fixed $T_{\rm rh}$ the state that reaches $\bar n_*$ gives a mass up to 44 per cent higher. `v6_state_at_bang.py` finds the transfer matrix of our half-crossing from the massless levels to the late adiabatic levels, and of the mirror's from the early adiabatic levels to the massless ones, and joins them with an exactly massless stretch. Over a period of the joining phase the in-vacuum occupation runs from zero, to the resolution of the phase grid, to $4n_{\rm vac}(1-n_{\rm vac})$, and the phase average of $(1-\sqrt{1-P})/2$ equals the elliptic formula to 10^{-3} at $x = 0.6$ and 1.2; dropping the factor $1-n_{\rm vac}$ fails the comparison. At fixed $t_{\rm rh}$ the bound halves production; at fixed $T_{\rm rh}$ the heavier mass also delays production, which is why the mass ratio exceeds $2^{2/5}$.

Energy at the bang. The assumption of Section 3.1 limits how far a state can move production through the energy its populations carry. With $p_e(x)$ the population of the upper massless level at the bang, measured from Boyle, Finn and Turok's Hadamard state, which on the dark-matter modes differs from the massless vacuum by $1/16x_b^4 \lesssim 10^{-25}$ and so leaves every entry below unchanged, the population that reaches $\bar n_*$ is $\sin^2[\frac12 \arctan(|\cos\theta|\tan 2\beta)]$ with $\sin^2\beta = n_{\rm vac}$, averaged over the interference phase θ. With phases tuned to the history, a population p_e at a mode whose interference phase has $|\cos\theta| = c$ changes the occupation by $\pm 2c\sqrt{p_e(1-p_e)n_{\rm vac}(1-n_{\rm vac})} + p_e(1-2n_{\rm vac})$, the upper sign raising it, and the least population that changes production by ten per cent chooses p_e for each mode and phase to maximise the change less a multiple of the energy. Only the second term survives for random phases, and the least population that raises production fills modes in order of $(1-2n_{\rm vac})/x$. Their energy relative to the plasma is $(30/\pi^4 g_*)\,\kappa^4 \int_0^\infty x^3 p_e\,dx$, with $\kappa = \sqrt{\pi M\sqrt{g_*/15}/\mu}\;e^N(\rho_{\rm end}/\rho_{\rm rh})^{1/12}$; the ratio is constant through the massless stretch, so it can be imposed at the onset of inflation as well as at the bang. A population that reaches $\bar n_*$ needs $\int x^3 p_e\,dx$ between 1.3×10^{-2} and 4.9×10^{-2}; one whose phases follow the later history needs 1.0×10^{-4} to 4.1×10^{-4} to change production by ten per cent either way, against 1.6×10^{-3} to 6.6×10^{-3} to raise it with random

phases. The energy goes as e^{4N}, so the largest change in production the condition allows falls as e^{-2N}: at 15 e-folds it is 0.1 to 1.8 per cent over these ranges. The e-folds above which each exceeds the plasma's energy at the bang, computed with a sudden end of reheating for $H_{\rm inf}$ from 10^{10} to 10^{13} GeV and $T_{\rm rh}$ of 1.87×10^{13} and 5×10^{13} GeV, are 11.5 to 12.9 for the state that reaches $\bar{n}_*$, 12.0 to 13.4 for random phases and 12.7 to 14.1 for tuned phases either way (Table A.1). At 8 e-folds none does, so the comparison can fail.

Table A.1. E-folds of inflation above which N_1 at the bang carries more energy than the plasma, for the state that reaches $\bar{n}_*$ and for the least populations that change production by ten per cent.

$H_{\rm inf}$ (GeV)	$T_{\rm rh}$ (GeV)	reaches $\bar{n}_*$	10% more, random phases	10% more, tuned phases	10% less, tuned phases
10^{10}	1.87×10^{13}	12.7	13.2	13.9	13.8
10^{12}	1.87×10^{13}	11.9	12.3	13.1	13.1
10^{13}	1.87×10^{13}	11.5	12.0	12.7	12.7
10^{10}	5×10^{13}	12.9	13.4	14.1	14.1
10^{12}	5×10^{13}	12.1	12.6	13.3	13.3
10^{13}	5×10^{13}	11.7	12.2	13.0	12.9

Code and data

Every number in this paper comes from one of these scripts, in Python 3.12 with numpy, scipy, mpmath and matplotlib: `v6_bath.py`, `v6_reheating_production.py`, `v6_mass_with_bath.py`, `v6_reheating_smooth.py`, `v6_mass_smooth.py`, `v6_hinf.py`, `v6_instant.py`, `v6_start_check.py`, `v6_hb_check.py`, `v6_state_at_bang.py`, `v6_quartic_decay.py`, `v6_trilinear_bath.py` and `v6_alpha_inflaton.py`, which builds the model of Section 4 and runs its checks. `fig_inflaton.py` draws the figures from their output, `mass_table.py` prints Table 1 from the mass tables the solvers write, and `inflaton_check.py` compares each number in the text with the script that produces it. Each script carries known-bad inputs that it must reject. All are in the code release of the companion paper [40], whose working repository is https://github.com/BenWiseman/separate-ways.

AI disclosure. The author originated the central ideas, directed the work, and is responsible for the writing, claims, and decisions in this paper. Anthropic's Claude Opus 5.5 was used under the author's direction to edit drafts, format references, write and revise the numerical and graphical code, search the literature, and run fact and consistency checks; MiniMax-M3 and DeepSeek were used as independent readers. These are large language models; nothing a model proposed was kept until it had been redone in a script or checked against a source, and no model is an author.

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